

KE-03: Locating a near-largest eigenvalue in the exact-query model

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Status of this manuscript: Independently checked resolution (Codex-agent review).

Outcome: Affirmative resolution in the displayed exact-query model.

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Target: [Repository entry KE-03](#), snapshot b412319.

The original proof draft was generated in a ChatGPT conversation. A separate Codex agent independently checked the complete proof against the exact repository target on 11 September 2026; its [detailed review](#) records **PASS**. The theorem and proof text below are unchanged from the reviewed submission. This verification is an independent agent review, not external human peer review or a formal proof certificate.

Theorem KE-03: proposed exact-query algorithm

Suppose $A \in \mathbb{C}^{n \times n}$ is diagonalizable,

$$A = V\Lambda V^{-1}, \quad \|V\|_2 \|V^{-1}\|_2 \leq K, \quad \rho = \rho(A) > 0,$$

where $K \geq 1$ is supplied but V and Λ are not. Only exact queries $v \mapsto Av$ are permitted.

Claim. For every $0 < \varepsilon < 1/2$, a randomized algorithm using

$$O(\varepsilon^{-2}[1 + \log(nK)])$$

queries returns $z \in \mathbb{C}$ with probability at least 0.997 such that some $\mu \in \sigma(A)$ satisfies

$$|\mu| \geq (1 - \varepsilon)\rho, \quad |z - \mu| \leq \varepsilon\rho.$$

All intervening work is finite exact arithmetic. Its bit complexity and floating-point stability are not bounded here.

1. A simultaneous polynomial estimate

Choose an integer $N \geq 1000n$ such that $N + 1$ is a power of two. Sample each coordinate of $b \in \mathbb{R}^n$ independently and uniformly from

$$\{0, 1/N, \dots, 1\}.$$

This uses finitely many random bits. Set $F = 1000n^2K$.

Lemma. With probability at least 0.997, simultaneously for every complex polynomial p ,

$$F^{-1} \max_{\lambda \in \sigma(A)} |p(\lambda)| \leq \|p(A)b\|_2 \leq F \max_{\lambda \in \sigma(A)} |p(\lambda)|.$$

Proof. Let w_i be row i of V^{-1} . Choose a coordinate j with $|(w_i)_j| \geq \|w_i\|_2/\sqrt{n}$. Conditional on the other coordinates of b , the event

$$|w_i b| < \frac{\|w_i\|_2}{1000n^2}$$

restricts the real variable b_j to an interval of length at most $2/(1000n^{3/2})$. This remains true for complex w_i : intersection of a disk with a line has diameter at most the disk's diameter.

An interval of length ℓ contains at most $N\ell + 1$ grid points. Thus the event has probability at most

$$\frac{2}{1000n^{3/2}} + \frac{1}{N+1} \leq \frac{3}{1000n}.$$

A union bound gives, with probability at least 0.997, for every i ,

$$|(V^{-1}b)_i| \geq \frac{\|w_i\|_2}{1000n^2} \geq \frac{1}{1000n^2\|V\|_2}.$$

The last inequality follows from $w_i V = e_i^T$. On this event, for every p ,

$$\begin{aligned} \|p(A)b\|_2 &\geq \frac{\|p(\Lambda)V^{-1}b\|_2}{\|V^{-1}\|_2} \\ &\geq \frac{\max_i |p(\lambda_i)|}{1000n^2\|V\|_2\|V^{-1}\|_2} \geq F^{-1} \max_i |p(\lambda_i)|. \end{aligned}$$

The deterministic upper bound is

$$\|p(A)b\|_2 \leq K\|b\|_2 \max_i |p(\lambda_i)| \leq K\sqrt{n} \max_i |p(\lambda_i)| \leq F \max_i |p(\lambda_i)|.$$

This proves the lemma. Its event depends only on b , so it also covers adaptively selected polynomials. $\square$

2. Query phase

Put

$$\eta = \frac{\varepsilon^2}{1024}, \quad m = \left\lceil \frac{\log F}{\log(1+\eta)} \right\rceil.$$

Use m queries to compute $b, Ab, \dots, A^m b$. For any $s \in \mathbb{C}$, the vector

$$(A + sI)^m b = \sum_{\ell=0}^m \binom{m}{\ell} s^{m-\ell} A^\ell b$$

can now be computed without further queries.

For analysis, write

$$q_s = \|(A + sI)^m b\|_2^{1/m}, \quad R_s = \rho(A + sI).$$

On the event of the lemma,

$$(1+\eta)^{-1}R_s \leq q_s \leq (1+\eta)R_s \quad \text{for every } s \in \mathbb{C}.$$

Compute a positive multiplicative approximation r to q_0 satisfying

$$(1+\eta)^{-1}q_0 \leq r \leq (1+\eta)q_0.$$

The finite-arithmetic implementation is given below. On the success event,

$$(1+\eta)^{-2}\rho \leq r \leq (1+\eta)^2\rho.$$

3. Selection among shifts

Choose a finite unit-circle net $\{u_\ell\}$ such that every unit complex number is within chordal distance $\varepsilon/8$ of some u_ℓ . A construction with $O(1/\varepsilon)$ rational points is given below.

Compute all squared norms $\|(A + ru_\ell I)^m b\|_2^2$, using the stored block powers. Choose any maximizing index $\hat{\ell}$, and return

$$z = ru_{\hat{\ell}}.$$

This selection requires no further oracle queries.

4. Location guarantee

Choose $\mu_* \in \sigma(A)$ with $|\mu_*| = \rho$. Some net point u_* obeys $|\mu_*/\rho - u_*| \leq \varepsilon/8$, so

$$R_{ru_*}^2 \geq |\mu_* + ru_*|^2 \geq (\rho + r)^2 - \frac{\rho r \varepsilon^2}{64}.$$

By the maximizing selection and the bounds on q_s ,

$$R_z \geq (1 + \eta)^{-2} R_{ru_*}.$$

Take $\mu \in \sigma(A)$ attaining $R_z = |\mu + z|$, and let $a = (1 + \eta)^{-4}$. Since $|\mu| \leq \rho$ and $|z| = r$,

$$\begin{aligned} |\mu - z|^2 &= 2(|\mu|^2 + r^2) - |\mu + z|^2 \\ &\leq (\rho - r)^2 + (1 - a)(\rho + r)^2 + a \frac{\rho r \varepsilon^2}{64}. \end{aligned}$$

The bounds on r imply $|\rho - r| \leq 3\eta\rho$ and $r \leq 2\rho$. Also $1 - a \leq 4\eta$. Consequently

$$|\mu - z|^2 \leq \left(9\eta^2 + 36\eta + \frac{\varepsilon^2}{32}\right) \rho^2 < \frac{\varepsilon^2 \rho^2}{9}.$$

For the final inequality, divide by ε^2 and use $\varepsilon^2 < 1/4$:

$$\frac{9\varepsilon^2}{1024^2} + \frac{36}{1024} + \frac{1}{32} < \frac{1}{9}.$$

Thus $|z - \mu| < \varepsilon\rho/3$, and

$$|\mu| \geq r - |z - \mu| \geq (1 - 2\eta - \varepsilon/3)\rho \geq (1 - \varepsilon)\rho.$$

Both requested guarantees follow on the same event of probability at least 0.997.

5. Finite exact arithmetic and query count

The displayed logarithms and roots need not be primitive exact operations. Find m by repeated multiplication and comparison until $(1 + \eta)^m \geq F$; this gives the same integer as the ceiling formula.

If $\|A^m b\|_2^2 = 0$, return zero immediately. This event is impossible on the success event because $\rho > 0$. Otherwise bracket q_0 with positive powers of two, comparing their $2m$ -th powers with $\|A^m b\|_2^2$. Starting from such a bracket $[L, U]$, bisect until $U \leq (1 + \eta)L$. Choosing $r = L$ gives the required multiplicative approximation. Bracketing and bisection terminate for every positive input in finite exact arithmetic.

For an exact unit-circle net, set

$$u(t) = \frac{1 - t^2 + 2it}{1 + t^2}, \quad -1 \leq t \leq 1.$$

Use $u(t)$ and $-u(t)$ on a rational uniform mesh of spacing at most $\varepsilon/8$ including both endpoints. Together the two arcs cover the circle. Since $|u'(t)| \leq 2$ and a point of the parameter interval is within half a mesh spacing of a mesh point, the required chordal error is at most $\varepsilon/8$. Such a mesh has $O(1/\varepsilon)$ points and uses rational arithmetic. Its size can be found by comparisons, without an exact ceiling primitive.

The shift selection compares squared norms directly; no m -th roots are needed. Complex arithmetic and squared norms can be implemented with arithmetic on real and imaginary parts. No multiplication by A^* and no shifted linear-system solve is used.

Finally, $\log F = O(1 + \log(nK))$ and $\log(1 + \eta) \geq \eta/2$, so

$$m = O(\varepsilon^{-2} [1 + \log(nK)]).$$

This proves the proposed query-complexity bound. Unrestricted intervening work is essential to the stated scope; no practical numerical implementation is certified. $\square$

Scope and review notes

The input is any diagonalizable complex A with positive spectral radius and a supplied eigenvector-condition bound K . Only queries $v \rightarrow Av$ are used. Intervening work is finite exact arithmetic, with unrestricted cost and precision, as allowed by the displayed target. No A^* oracle or shifted-system oracle is used. This is not a floating-point stability, bit-complexity, or total-runtime result.

The independent review checked simultaneous anti-concentration for complex eigenvector coordinates, adaptive use of the same Krylov sequence, shift-selection geometry, and finite-arithmetic termination. The supplied diagnostic code checks only the shift geometry, not an end-to-end stable implementation of the high-degree algorithm.

The packaging pass checked the repository statement and contribution rules on 11 September 2026. It did not conduct a new exhaustive literature search or establish novelty.

Source references

Alex Townsend, *Open Problems in Numerical Linear Algebra* (2026), [entry KE-03](#), repository snapshot b412319, accessed 11 September 2026.

Amsel et al., workshop report, [arXiv:2602.05394v3](#), Problem 3.9. The exact oracle formulation with supplied K is the displayed repository target; the present claim is explicitly limited to that formulation rather than every possible interpretation of the workshop question.